

Task 7: Sensor Data Analysis with NumPy, Pandas & Visualization



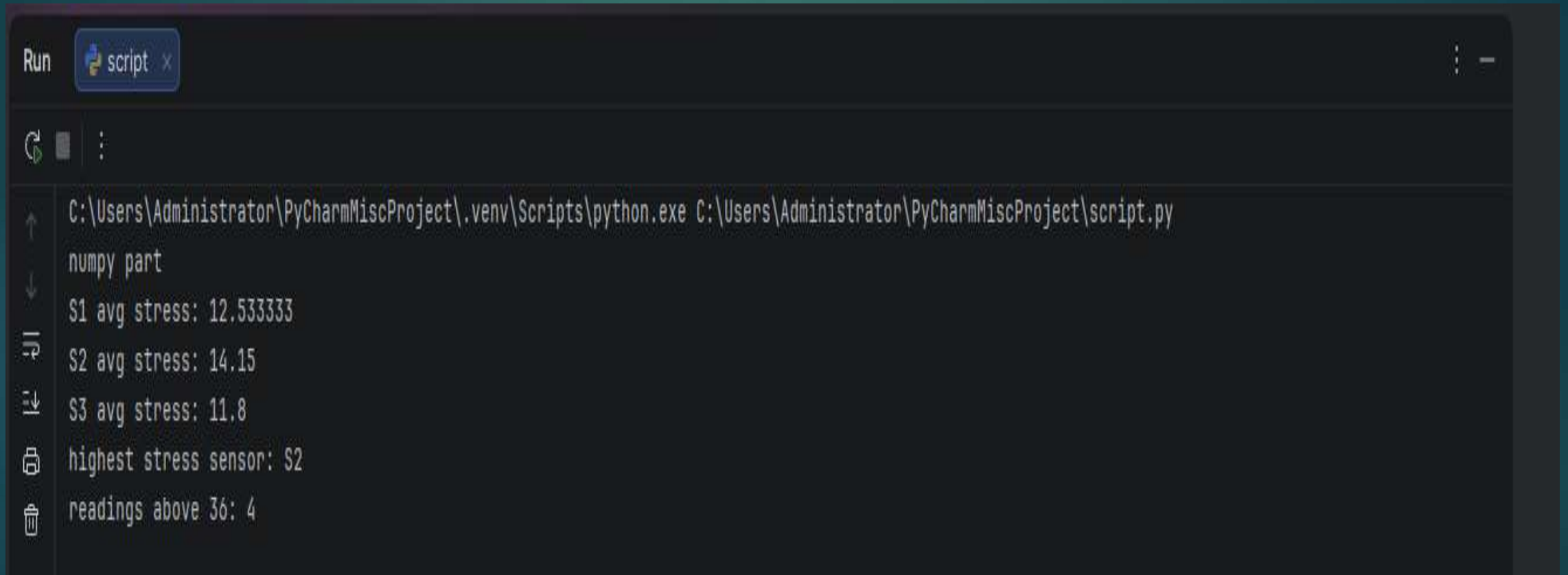
Part 1: NumPy Operations

► Code :

```
14 # numpy array
15 my_data = np.array(sensor_records, dtype=[('id', 'U5'), ('time', 'U20'), ('temp', 'F4'), ('stress', 'F4')])
16
17 sensors = np.unique(my_data['id'])
18
19 print("numpy part")
20 for x in sensors:
21     one_sensor = my_data[my_data['id'] == x]
22     avg_stress = np.mean(one_sensor['stress'])
23     print(x, "avg stress:", avg_stress)
24
25 # find highest per sensor
26 high_one = ""
27 high_num = 0
28 for x in sensors:
29     one_sensor = my_data[my_data['id'] == x]
30     avg_s = np.mean(one_sensor['stress'])
31     if avg_s > high_num:
32         high_num = avg_s
33         high_one = x
34 print("highest stress sensor:", high_one)
35
36 # temps above 36
37 hot_readings = my_data[my_data['temp'] > 36]
38 print("readings above 36:", len(hot_readings))
39
```



Code Results



The image shows a PyCharm Run console window. At the top, there is a 'Run' button and a tab labeled 'script'. Below the tab, there are icons for running, debugging, and a vertical ellipsis. The main area of the console displays the output of a Python script. The first line is the command being executed: 'C:\Users\Administrator\PyCharmMiscProject\.venv\Scripts\python.exe C:\Users\Administrator\PyCharmMiscProject\script.py'. The subsequent lines are the output of the script: 'numpy part', 'S1 avg stress: 12.533333', 'S2 avg stress: 14.15', 'S3 avg stress: 11.8', 'highest stress sensor: S2', and 'readings above 36: 4'. On the left side of the console, there are icons for navigating through the output: up and down arrows, a list icon, a search icon, a copy icon, and a delete icon.

```
C:\Users\Administrator\PyCharmMiscProject\.venv\Scripts\python.exe C:\Users\Administrator\PyCharmMiscProject\script.py
numpy part
S1 avg stress: 12.533333
S2 avg stress: 14.15
S3 avg stress: 11.8
highest stress sensor: S2
readings above 36: 4
```

Part 2: Pandas Operations Part

► Code:

```
Current File ▾ ▶ ⚙️ ⋮ @ 👤 🔍 ⚙️ Unlock Pro - 📄
Welcome to PyCharm script.py Task 7 Numpy, Pandas and Matplotlib visuals.py ✕
40 print("\npandas part")
41
42 # dataframe
43 df = pd.DataFrame(sensor_records, columns=['id', 'time', 'temp', 'stress', 'displacement'])
44 df['time'] = pd.to_datetime(df['time'])
45
46 # averages per sensor
47 avg_table = df.groupby('id')[['temp', 'stress', 'displacement']].mean()
48 print(avg_table)
49
50 # hottest
51 temp_avg = df.groupby('id')['temp'].mean()
52 hottest_one = temp_avg.idxmax()
53 print("hottest sensor:", hottest_one)
54
55 # visualize temp over time
56 plt.figure()
57 for sid in df['id'].unique():
58     small = df[df['id'] == sid]
59     plt.plot(small['time'], small['temp'], marker='o', label=sid)
60
61 plt.title("temperature over time")
62 plt.xlabel("time")
63 plt.ylabel("temp (C)")
64 plt.legend()
65 plt.grid(True)
```

Task 7 Numpy, Pandas and Matplotlib visuals.p 25:1 CRLF UTF-8 4 spaces Python 3.14 (PyCharmMiscProject)



Code Results

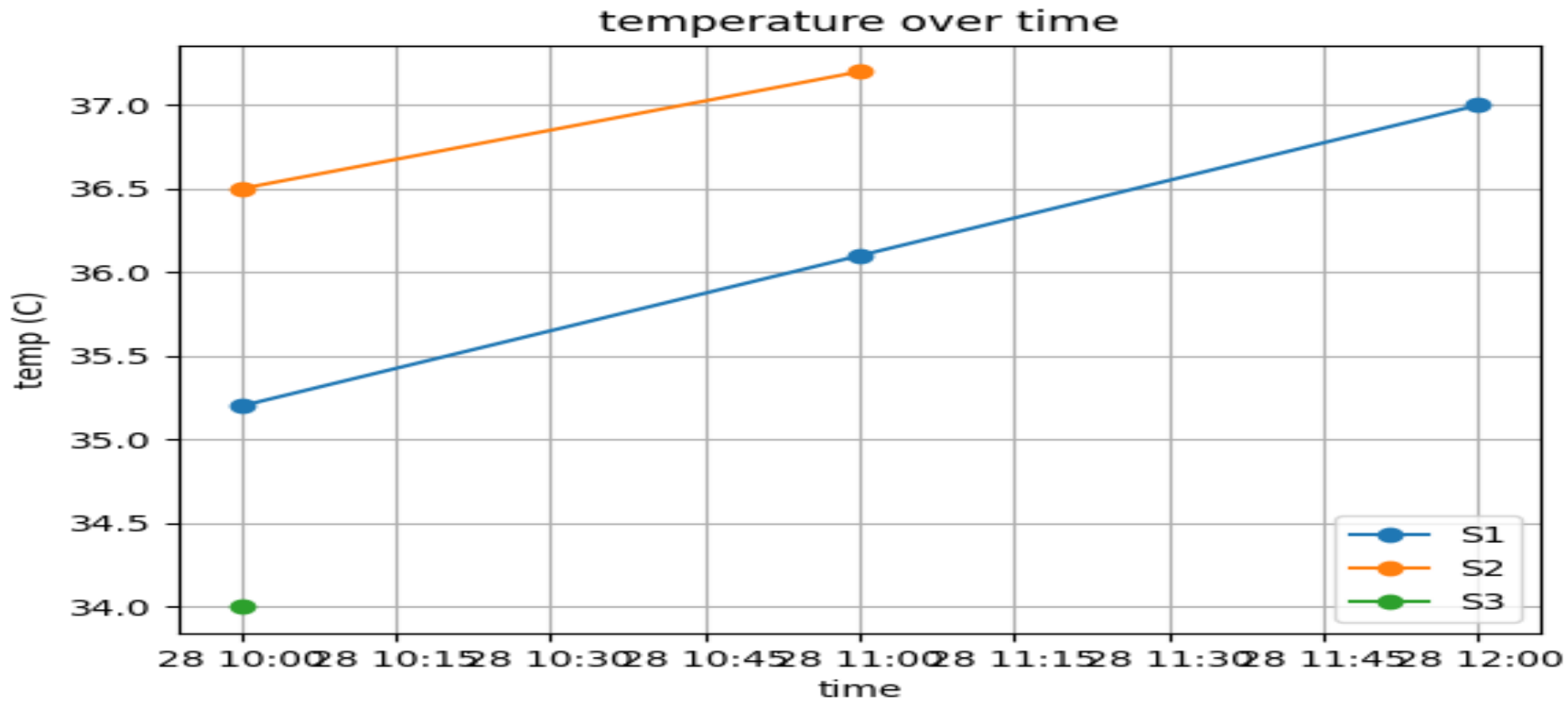
```
pandas part
      temp  stress displacement
id
S1  36.10  12.533333      0.00210
S2  36.85  14.150000      0.00305
S3  34.00  11.800000      0.00250
hottest sensor: S2

Process finished with exit code 0
|
```

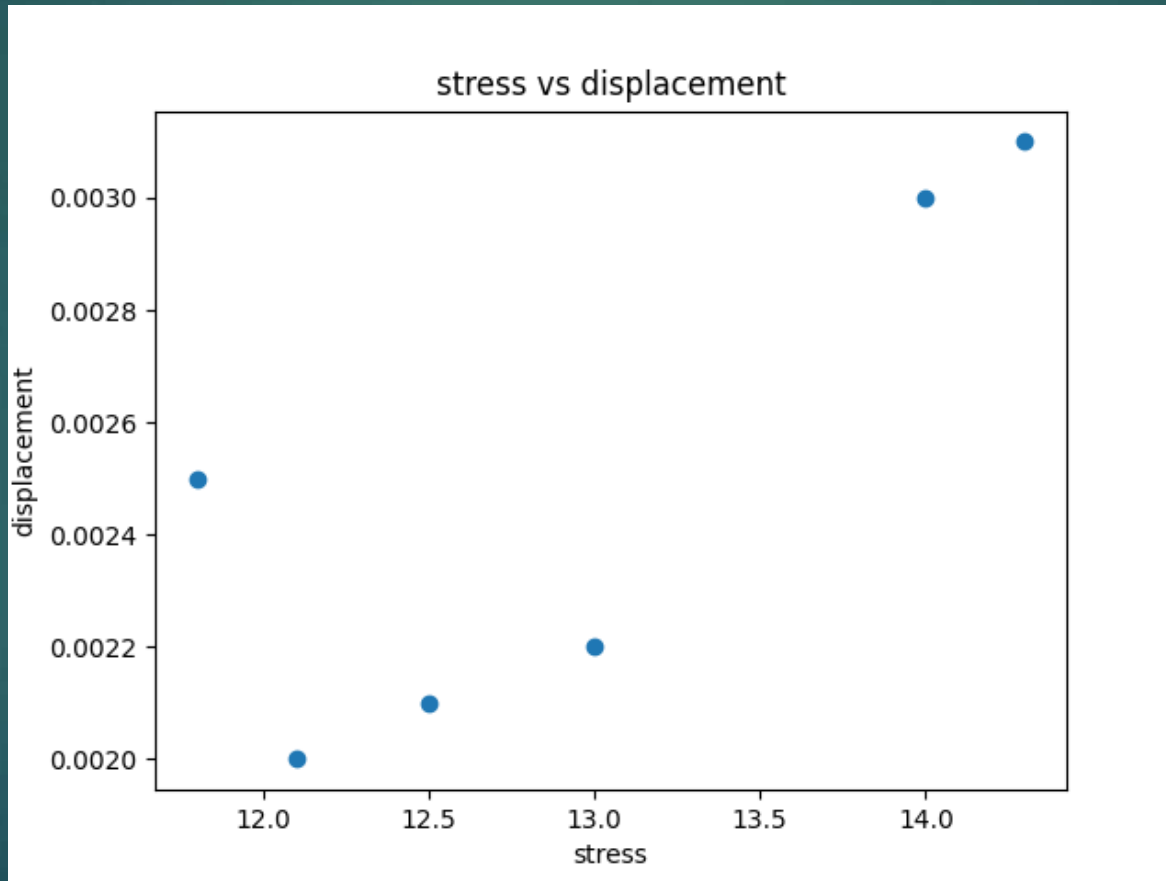
Desktop > New folder > python > Task 7, Numpy, Pandas & Visuals > Task 7 Numpy, Pandas and Matplotlib visuals.p 25:1 CRLF UTF-8 4 spaces Python 3.14 (PyCharmMiscProject)

PART 3: Visualization Part

Line Chart



Scattered Plot



Conclusion

- ▶ **NumPy** : Array and Filtering.

- ▶ **Pandas** : Data Analysis.

- ▶ **Visualization:**

- 1- Line chart to show relationship between temperature and time.

- 2- Scattered plot to show relationship between stress and displacement.

